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Symplectic or contact structures on Lie groups

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Abstract

We study left invariant contact forms and left invariant symplectic forms on Lie groups. In the case of filiform Lie groups we give a necessary and sufficient condition for the existence of a left invariant contact form and we prove the uniqueness of this contact form up to a nonzero scalar multiple. As an application we classify all symplectic structures on nilpotent Lie algebras of dimension $\leqslant 6$.

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Introduction

In the sequel G denotes a Lie group (supposed to be connected for simplicity) with Lie algebra $\mathfrak{g}$. If G is endowed with a left invariant differential 1-form α^+ such that

$$\alpha^+ \wedge (d\alpha^+)^p \neq 0$$

where $2p+1$ is the dimension of G, we will say that the pair (G, α^+) is a contact Lie group and that $(\mathfrak{g}, \alpha)$ is a contact Lie algebra; here $\alpha := \alpha^+_\varepsilon$ (ε is the unit element of G).

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Following Lichnerowicz and Medina [20,21] a pair (G, Ω^+) where Ω^+ is a left invariant symplectic form, is called a symplectic Lie group and the corresponding infinitesimal object $(\mathfrak{g}, \omega)$, where $\omega := \Omega^+_\varepsilon$, is called a symplectic Lie algebra.

In [22] (see also [6,7]) a method of construction of symplectic Lie algebras, called "symplectic double extension", is described. According to in [22, Theorem 2.5] every nilpotent symplectic Lie algebra is obtained from a sequence of "symplectic double extensions" starting from the trivial abelian Lie algebra consisting of only one element.

This result immediately implies that every nilpotent contact Lie algebra can be obtained by two operations, namely: the "symplectic double extension" and the contactization.

Here we give a little explanation about the main results and the organization of this work.

Section 1 gives a geometric description of contact Lie groups.

Section 2 supplies a necessary and sufficient condition for a filiform Lie group to possess a left invariant contact form (see Theorem 3). Such a contact form is unique up to a nonzero scalar multiple and has a simple expression in terms of an adapted basis (see Theorem 4).

Here it is convenient to recall some known facts. According to Gromov's result, every Lie group of odd dimension admits a nonnecessary left invariant contact form. A symplectic Lie group (G, Ω^+) is endowed with a left invariant affine structure (see [5]), that is a flat torsionless linear connection ∇, defined by

$$\omega(ab, c) = -\omega\big(b, [a, c]\big), \quad a, b, c \in \mathfrak{g},$$
$$\nabla_{a^+} b^+ := (ab)^+,$$

where a^+ is the left invariant vector field on G such that $a^+_\varepsilon = a$. Such a connection ∇ is fundamental in the description of symplectic Lie groups and specially Kählerian Lie groups (see [6,7]). Unlike the symplectic case there exist contact Lie groups with no left invariant affine structure. This happens for semisimple contact Lie groups. More surprising, there even exist nilpotent contact Lie groups that never admit such an affine structure: a direct verification allows us to check that the example of Benoist (of dimension 11) supplied in [3] is among them.

Our work ends by the classification of all nilpotent symplectic Lie algebras of dimension $\leqslant 6$.

All Lie algebras are real unless explicitly mentioned.

In this paper the following standard convention will be used without explicit mentioning: for a concrete basis $X_1, \ldots, X_n$ of a Lie algebra only nonzero brackets $[X_i, X_j]$, $i < j$, will be explicitly indicated.

1. Contact Lie groups as principal bundles with connection

The aim of this section is to prove the following more or less known results (see [1,4,12,13,16,19]).

Theorem 1. *Let G be a connected Lie group. Then the left invariant contact forms α^+ on G are in bijection with the coadjoint orbits $\mathrm{Orb}(\alpha) \equiv G/H$ of codimension one. These orbits are reductive homogeneous manifolds and α^+ is a connection form in the principal bundle $G(G/H, H)$ with curvature form the pullback $\widetilde{\Omega}$ of the Kirilov–Kostant–Souriau symplectic form Ω on G/H. Furthermore if H_0 is the connected component of the unit of H, $p : G/H_0 \to G/H$ the canonical map and $\Omega_0 = p^*(\Omega)$, then the canonical action of G on $(G/H_0, \Omega_0)$ is Hamiltonian.*

Proof. Consider a left invariant contact form α^+ on G where $\dim G = 2p+1$. As $\mathrm{Ker}\,\alpha \cap \mathfrak{h} = \{0\}$ (where $\mathfrak{h}$ is the Lie algebra of H and $\alpha = \alpha^+_\varepsilon$) we have $0 \leqslant \dim(\mathrm{Ker}\,\alpha + \mathfrak{h}) = \dim \mathrm{Ker}\,\alpha + \dim \mathfrak{h} = 2p + \dim \mathfrak{h} \leqslant 2p+1$, and this implies $\dim \mathfrak{h} \leqslant 1$. On the other hand, since $\mathrm{Ad}(H) \ker \alpha = \ker \alpha$ the homogeneous space G/H is reductive. In this case, the $\ker\alpha$-component θ of the Maurer–Cartan form on G, with respect to the decomposition $\mathfrak{g} = \mathfrak{h} \oplus \mathrm{Ker}\,\alpha$ defines a connection in the bundle $G(G/H, H)$ which is invariant by the translations of G. This component coincides with α^+.

Moreover the left G-invariant form $\widetilde{\Omega} = d\alpha^+$ is also right H-invariant with kernel $\mathfrak{h}$. It determines the Kirilov–Kostant–Souriau symplectic form Ω on G/H. So, $\widetilde{\Omega} = \pi^*(\Omega)$ where $\pi : G \to G/H$ is the canonical projection and $\dim \mathfrak{h} = 1$.

The formula

$$d\alpha^+(X,Y) = -\frac{1}{2}\big[\alpha^+(X), \alpha^+(Y)\big] + \widetilde{\Omega}(X,Y)$$

for all $X, Y \in T_\sigma(G)$, shows that $\widetilde{\Omega}$ is the curvature form of α^+.

Let H_0 and $\rho : G/H_0 =: M_0 \to G/H$ be the connected component of the identity of H and the canonical covering map, respectively. It is clear that $\Omega_0 := \rho^*(\Omega)$ is symplectic and invariant by the canonical action Φ of G on M_0. We have also $\pi_0^*(\Omega_0) = \widetilde{\Omega}$ where $\pi_0 : G \to M_0$ is the canonical map.

We must prove that Φ is a Hamiltonian action.

For $x \in \mathfrak{g}$ let $[\tilde{x}]$ be the fundamental vector field on M_0 associated to x and $\tilde{x}$ its horizontal lift. Since the flow of $\tilde{x}$ is mapped on the flow of $[\tilde{x}]$ by π and

$$[\tilde{x}]_{[\sigma]} := \frac{d}{dt}\bigg|_{t=0} \exp tx \cdot [\sigma] = \frac{d}{dt}\bigg|_{t=0} [\exp tx \cdot \sigma]$$

it follows that the flow of $\tilde{x}$ consists of left translations on G. Thus $\tilde{x}$ is a right invariant vector field on G, i.e., $\tilde{x} = y^-$ for some $y \in \mathfrak{g}$ satisfying $(\pi_0)_{*,\varepsilon}(y) = [x]_{\pi_0(\varepsilon)}$. Let $(\varphi_t)_t$ be the flow of y^-. One has,

$$\begin{aligned} 0 = \mathcal{L}(y^-)\alpha^+ &= \big(d \circ i(y^-) + i(y^-) \circ d\big)(\alpha^+) \\ &= d\big(\alpha^+(y^-)\big) + \big(i(y^-) \circ d\big)(\alpha^+) \end{aligned}$$

where $\mathcal{L}$ is the Lie derivative.

Moreover

$$i(y^-)\, d\alpha^+ = i(y^-)\pi_0^*(\Omega_0) = \pi_0^*\big(i[\tilde{x}]\big)(\Omega_0)$$

so that

$$0 = d\big(\alpha^+(y^-)\big) + \pi_0^*\big(i[\tilde{x}]\Omega_0\big).$$

We shall prove that the function $f_y : G \to \mathbb{R}$ given by, $f_y(\sigma) := \alpha^+_\sigma(y^-_\sigma)$ can be projected by π_0.

Let $Y \in T_\sigma(G)$. From the last above identity we have

$$d\big(\alpha^+(y^-)\big)(Y_\sigma) = Y_\sigma\big(\alpha^+(y^-)\big) = Y_\sigma(f_y) = -\Omega_0\big([\tilde{x}], (\pi_0)_{*,\sigma} Y_\sigma\big).$$

Consequently if Y is tangent to the fiber i.e., if $(\pi_0)_{*,\sigma} Y_\sigma = 0$, we'll have $Y_\sigma(f_y) = 0$ for every σ in G. Hence f is constant along H_0. This implies the existence of a smooth function $J_y \circ \pi_0 = f_y$. $\square$

Since the center $Z(G)$ has dimension not bigger then 1, we get the following corollary.

Corollary 1 [12].

(a) *If* (G,α^+) *is a contact Lie group of nondiscrete center* $Z(G)$*, then the factor group* $G/Z(G)$ *is endowed with a left invariant symplectic form* Ω^+ *such that* $\pi^*\Omega^+ = -d\alpha^+$*, where* $\pi : G \to G/Z(G)$ *is the canonical projection.*

(b) *Conversely, if* (K,Ω^+) *is a symplectic Lie group with Lie algebra* $\mathfrak{k}$*, every Lie group* G *with Lie algebra the central extension of* $\mathfrak{k}$ *by* $\mathbb{R}$ *via* $\omega := \Omega^+_\varepsilon$ *admits a left invariant contact form* α^+ *satisfying* $\pi^*\Omega^+ = -d\alpha^+$ (*that is* $\pi^*\omega = -d\alpha$).

Remark 1. If two contact Lie algebras $(\mathfrak{g}_1,\alpha_1)$ and $(\mathfrak{g}_2,\alpha_2)$ with nontrivial centers are contacto-isomorphic (that is there exists an isomorphism of Lie algebras $\varphi : \mathfrak{g}_1 \to \mathfrak{g}_2$, such that, $\varphi^*(\alpha_2) = \alpha_1$), then the symplectic Lie algebras $(\mathfrak{g}_1/Z(\mathfrak{g}_1),\omega_1)$ and $(\mathfrak{g}_2/Z(\mathfrak{g}_2),\omega_2)$ are symplecto-isomorphic (that is there exists an isomorphism of Lie algebras $\varphi : \mathfrak{g}_1/Z(\mathfrak{g}_1) \to \mathfrak{g}_2/Z(\mathfrak{g}_2)$, such that, $\varphi^*(\omega_2) = \omega_1$) and $\pi_i^*\omega_i = -d\alpha_i$, $i = 1,2$; where $\pi_i : \mathfrak{g}_i \to \mathfrak{g}_i/Z(\mathfrak{g}_i)$ are the canonical projections.

2. Contact or symplectic filiform Lie algebras

2.1. Filiform Lie algebras (basic definitions and results)

Let $\mathfrak{g}$ be a nilpotent Lie algebra of dimension m. Let

$$C^0\mathfrak{g} \supset C^1\mathfrak{g} \supset \cdots \supset C^{m-2}\mathfrak{g} \supset C^{m-1}\mathfrak{g} = \{0\}$$

be the descending central series of $\mathfrak{g}$, where $C^0\mathfrak{g} = \mathfrak{g}$, $C^i\mathfrak{g} = [\mathfrak{g}, C^{i-1}\mathfrak{g}]$, $1 \leqslant i \leqslant m-1$.

Definition 1. A Lie algebra $\mathfrak{g}$ of dimension $\geqslant 3$ is called *filiform* if $\dim C^k\mathfrak{g} = m-k-1$ for $k = 1,\ldots,m-1$. A Lie group G is called *filiform* if its Lie algebra $\mathfrak{g}$ is filiform.

We remark that the filiform Lie algebras have the maximal possible nilindex, that is $m-1$. These algebras are the “least” nilpotent.

2.1.1. Examples of filiform Lie algebras

For each $n \in \mathbb{N}$ there exists several $(n+1)$-dimensional filiform Lie algebras which are particularly interesting. In the following description, the brackets are given relative to a basis $(X_0, X_1, \ldots, X_n)$.

1. The Lie algebra L_n:
   It is the simplest $(n+1)$-dimensional filiform Lie algebra. Its nontrivial brackets are given by:
   $$[X_0, X_i] = X_{i+1}, \quad i = 1,\ldots,n-1.$$
2. The Lie algebra Q_n $(n = 2k+1)$:
   $$[X_0, X_i] = X_{i+1}, \quad i = 1,\ldots,n-1,$$
   $$[X_i, X_{n-i}] = (-1)^i X_n, \quad i = 1,\ldots,k.$$
   In the basis $(Z_0, Z_1, \ldots, Z_n)$, where $Z_0 = X_0 + X_1$, $Z_i = X_i$, $i = 1,\ldots,n$; this Lie algebra is defined by
   $$[Z_0, Z_i] = Z_{i+1}, \quad i = 1,\ldots,n-2,$$
   $$[Z_i, Z_{n-i}] = (-1)^i Z_n, \quad i = 1,\ldots,k.$$

Let $\mathfrak{g}$ be a m-dimensional filiform Lie algebra. It is naturally filtered by its descending central series, and we can associate to $\mathfrak{g}$ a graded Lie algebra $gr\mathfrak{g}$ which is also filiform. This Lie algebra is defined on the vector space

$$gr\mathfrak{g} = \bigoplus_{i=1}^{m-1} \mathfrak{g}_i$$

where $\mathfrak{g}_i = C^{i-1}\mathfrak{g}/C^i\mathfrak{g}$, by the brackets $[x + C^i\mathfrak{g}, y + C^j\mathfrak{g}] = [x, y] + C^{i+j}\mathfrak{g}$, $x \in C^{i-1}\mathfrak{g}$, $y \in C^{j-1}\mathfrak{g}$.

Proposition 1 [26]. *Let $\mathfrak{g}$ be a m-dimensional filiform Lie algebra. Then the graded Lie algebra $gr\mathfrak{g}$ is isomorphic to L_{m-1}, if m is odd, and isomorphic to L_{m-1} or Q_{m-1}, if m is even.*

Let Δ be the set of pairs of integers (k, r) such that $1 \leqslant k \leqslant n - 1$, $2k + 1 < r \leqslant n$, $r \geqslant 4$ (if n is odd we suppose that Δ contains also the pair $(\frac{n-1}{2}, n)$). For any element $(k, r) \in \Delta$, we can associate the following 2-cocycle for the Chevalley cohomology of L_n with coefficients in the adjoint module denoted by $\Psi_{k,r}$ and defined by

$$\Psi_{k,r}(X_i, X_j) = -\Psi_{k,r}(X_j, X_i) = (-1)^{k-i} C_{j-k-1}^{k-i} X_{i+j+r-2k-1}$$

if $1 \leqslant i \leqslant k < j \leqslant n$, $i + j + r - 2k - 1 \leqslant n$ and $\Psi_{k,r}(X_i, X_j) = 0$ otherwise. We remark that this formula for $\Psi_{k,r}$ is uniquely determined by the following conditions:

$$\Psi_{k,r}(X_k, X_{k+1}) = X_r,$$
$$\Psi_{k,r}(X_i, X_j) \in Z^2(L_n, L_n).$$

Proposition 2 [26]. *Any $(n + 1)$-dimensional filiform Lie algebra law μ is isomorphic to $\mu_0 + \Psi$ where μ_0 is the law of L_n and Ψ is a 2-cocycle defined by*

$$\Psi = \sum_{(k,r)\in\Delta} a_{k,r}\Psi_{k,r}$$

and verifying the relation $\Psi \circ \Psi = 0$ with

$$\Psi \circ \Psi(x, y, z) = \Psi\bigl(\Psi(x, y), z\bigr) + \Psi\bigl(\Psi(y, z), x\bigr) + \Psi\bigl(\Psi(z, x), y\bigr).$$

Conversely, for all sets $\{a_{k,r}\colon (k, r) \in \Delta,\ a_{k,r} \in \mathbb{R}\}$ such that $\Psi \circ \Psi = 0$ the law $\mu_0 + \Psi$ is an $(n + 1)$-dimensional filiform Lie algebra law.

Remark 2. A detailed study of the cocycles Ψ verifying the relation $\Psi \circ \Psi = 0$ is given in [9,14,17,18].

Corollary 2. *Any $(n + 1)$-dimensional filiform Lie algebra is defined in a basis $(X_0, X_1, \ldots, X_n)$ by the brackets*

$$[X_0, X_i] = X_{i+1}, \quad i = 1, \ldots, n - 1,$$
$$[X_k, X_{k+1}] = \sum_{r=2k+2}^{2p} a_{k,r}X_r, \quad k = 1, \ldots, \left[\frac{n-2}{2}\right];\ a_{k,r} \in \mathbb{R},$$
$$\bigl[X_{\frac{n-1}{2}}, X_{\frac{n+1}{2}}\bigr] = a_{\frac{n-1}{2},n}X_n \quad (\textit{if } n \textit{ is odd}).$$

Definition 2. Let $\mathfrak{g}$ be a $(n+1)$-dimensional filiform Lie algebra with law μ. A basis $(X_0, X_1, \ldots, X_n)$ of $\mathfrak{g}$ is called *adapted,* if $[X_i, X_j] = \mu_0(X_i, X_j) + \Psi(X_i, X_j),\ 0 \leqslant i, j \leqslant n$, where μ_0 is the law of L_n.

Proposition 3. *Let* $\mathfrak{g}$ *be a filiform Lie algebra of dimension* $\geqslant 4$. *Then* $\operatorname{Der}\mathfrak{g}$ *is solvable.*

Proof. Consider an adapted basis $(X_0, X_1, \ldots, X_n)$ of $\mathfrak{g}$. As the central descending series of $\mathfrak{g}$ is an invariant flag under all derivations it is sufficient to show that the ideal $\langle X_1, \ldots, X_n\rangle$ is also an invariant. Let $d \in \operatorname{Der}\mathfrak{g}$ and $d(X_1) = \sum_{i=0}^{n} a_i X_i$. For the 4-dimensional filiform Lie algebra $\bar{\mathfrak{g}} = \mathfrak{g}/C^3\mathfrak{g}$ (it is isomorphic to L_3) we have the derivation $\overline{d}$ with $\overline{d}(\overline{X}_1) = \sum_{i=0}^{3} a_i \overline{X}_i$. This is possible only if $a_0 = 0$. $\square$

Let $\mathfrak{g}$ be a Lie algebra. Consider in a maximally abelian subalgebra $\mathfrak{t}$ of $\operatorname{Der}\mathfrak{g}$ consisting of semisimple endomorphisms (such a subalgebra is called torus of $\mathfrak{g}$). According to a theorem by Mostow [23] two such subalgebras are conjugated by an inner automorphism. The common dimension of all tori of $\mathfrak{g}$ is called the *rank* of $\mathfrak{g}$. Note that for nilpotent $\mathfrak{g}$ the rank cannot exceed the codimension of the derived ideal since $\mathfrak{g}$ is generated by any vector subspace of $\mathfrak{g}$ complementary to the derived ideal. For a filiform Lie algebra the only possible ranks are 0, 1 and 2.

Proposition 4 [15]. *Let* $\mathfrak{g}$ *be a filiform Lie algebra of dimension* $n+1$ *and of rank* 2. *Then* $\mathfrak{g}$ *is isomorphic to* L_n *if* n *is even and isomorphic to* L_n *or* Q_n *if* n *is odd.*

A description of the filiform Lie algebras of rank 0 and 1 is given in [15] (see also [18]).

2.2. Symplectization and contactization of the filiform Lie groups

The following result shows that the class of filiform Lie groups is closed with respect to the contactization and symplectization process described in the Section 1 and the study of the symplectic filiform Lie groups is equivalent to study of the contact filiform Lie groups.

Theorem 2. *Let* (G, α^+) *be a contact filiform Lie group. Then the quotient* $G/Z(G)$ *is a symplectic filiform Lie group. Conversely, if* (K, Ω^+) *is a symplectic filiform Lie group, every Lie group* G *with Lie algebra the central extension of* $\mathfrak{k}$ *by* $\mathbb{R}$ *via* $\omega := \Omega_\varepsilon^+$ *admits a left invariant contact form* α^+ *satisfying* $\pi^*\Omega^+ = -d\alpha^+$ *(that is* $\pi^*\omega = -d\alpha$*).*

Proof. Let $\mathfrak{g}$ be the Lie algebra of G and α its contact form. For an adapted basis $(X_0, X_1, \ldots, X_{2p})$ of $\mathfrak{g}$ we have $[X_0, X_i] = X_{i+1}$, $i = 1, \ldots, 2p-1$, and $Z(\mathfrak{g}) = \mathbb{R}\cdot X_{2p}$. Let $\pi : \mathfrak{g} \to \tilde{\mathfrak{g}} = \mathfrak{g}/Z(\mathfrak{g})$ be the canonical projection. Then we have

$$\bigl[\pi(X_0), \pi(X_i)\bigr] = \pi(X_{i+1}), \quad i = 1, \ldots, 2p-2;$$

and the Lie algebra $\tilde{\mathfrak{g}}$ is also filiform. According to the corollary of Theorem 1 the quotient $\mathfrak{g}/Z(\mathfrak{g})$ is a symplectic Lie algebra with a symplectic form ω satisfying $\pi^*\omega = -d\alpha$.

Conversely suppose that $(\tilde{\mathfrak{g}}, \omega)$ is a symplectic filiform Lie algebra of dimension $2p$. Consider an adapted basis $(Y_0, Y_1, \ldots, Y_{2p-1})$ of $\tilde{\mathfrak{g}}$. As ω is a nondegenerate form, there exists $0 \leqslant k \leqslant 2p-2$, such that $\omega(Y_k, Y_{2p-1}) \neq 0$. Let $\mathfrak{g} = \tilde{\mathfrak{g}} \oplus_\omega \mathbb{R}$ be the central extension defined by ω. The central descending sequence $\{C^i\mathfrak{g}\}$ satisfies the condition $\dim C^{i-1}\mathfrak{g}/C^i\mathfrak{g} = 1$ for all $2 \leqslant i \leqslant 2p-2$ because this property holds in $\tilde{\mathfrak{g}}$. As $\omega(Y_k, Y_{2p-1}) \neq 0$ we have also $\dim C^{2p-2}\mathfrak{g}/C^{2p-1}\mathfrak{g} = 1$ and $\dim C^{2p-1}\mathfrak{g} = 1$. Thus the nilindex of $\mathfrak{g}$ is equal to $2p$ and $\mathfrak{g}$ is filiform. According to the corollary of Theorem 1 the filiform Lie algebra $\mathfrak{g}$ admits a contact form α such that $\pi^*\omega = -d\alpha$, where π is the canonical projection $\pi : \mathfrak{g} \to \mathfrak{g}/Z(\mathfrak{g}) \cong \tilde{\mathfrak{g}}$. $\square$

2.3. Existence of a left invariant contact form

According to Proposition 2 and its corollary an arbitrary $(2p+1)$-dimensional filiform Lie algebra can be described in a basis $(X_0, X_1, \dots, X_{2p})$ by the law $\mu_0 + \Psi$ where μ_0 is the law of L_n and Ψ is a 2-cocycle defined by

$$\Psi = \sum_{(k,r)\in\Delta} a_{k,r}\Psi_{k,r}$$

and conversely, for all set $\{a_{k,r}\colon (k,r) \in \Delta,\ a_{k,r} \in \mathbb{R}\}$ such that $\Psi \circ \Psi = 0$ the law $\mu_0 + \Psi$ is an $(n+1)$-dimensional filiform Lie algebra law. The law $\mu_0 + \Psi$ can be also given by the following brackets:

$$\begin{aligned} &[X_0, X_i] = X_{i+1}, \quad i = 1, \dots, 2p-1, \\ &[X_k, X_{k+1}] = \sum_{r=2k+2}^{2p} a_{k,r} X_r, \quad k = 1, \dots, p-1. \end{aligned} \tag{$*$}$$

Theorem 3. *Let G be a $(2p+1)$-dimensional filiform Lie group, $\mathfrak{g}$ its Lie algebra defined in the basis $(X_0, X_1, \dots, X_{2p})$ by the brackets $(*)$, and let $(\alpha_0, \alpha_1, \dots, \alpha_{2p})$ be the dual basis. Then G admits a left invariant contact form if and only if*

$$A_j := \sum_{s=0}^{j-1} (-1)^s a_{p-j+s,2p-2(j-s-1)} C^s_{2j-s-2} \neq 0, \quad j = 1, 2, \dots, p-1.$$

If this property holds the linear form $\alpha = b_0\alpha_0 + b_1\alpha_1 + \cdots + b_{2p}\alpha_{2p}$, where $(\alpha_0, \alpha_1, \dots, \alpha_{2p})$ is the dual basis to $(X_0, X_1, \dots, X_{2p})$, is a contact form on $\mathfrak{g}$ if and only if $b_{2p} \neq 0$.

Proof. Let $A_j \neq 0$, $j = 1, 2, \dots, p-1$. We have

$$\begin{aligned} &[X_0, X_{2p-1}] = X_{2p}, \\ &[X_1, X_{2p-2}] = \sum_{k=1}^{p-1} a_{k,2k+2}\Psi_{k,2k+2}(X_1, X_{2p-2}) = A_{p-1}X_{2p}, \\ &[X_2, X_{2p-3}] = \sum_{k=2}^{p-1} a_{k,2k+2}\Psi_{k,2k+2}(X_2, X_{2p-3}) = A_{p-2}X_{2p}, \\ &\dots \\ &[X_{p-1}, X_p] = a_{p-1,2p}\Psi_{p-1,2p}(X_{p-1}, X_p) = A_1 X_{2p} \end{aligned}$$

and $[X_j, X_m] = 0$, if $m \geqslant 2p-j$. In the dual basis $(\alpha_0, \alpha_1, \dots, \alpha_{2p})$ of $\mathfrak{g}^*$ the previous brackets give

$$\begin{aligned} d\alpha = {} & -\alpha_0 \wedge \alpha_{2p-1} - A_{p-1}\alpha_1 \wedge \alpha_{2p-2} - A_{p-2}\alpha_2 \wedge \alpha_{2p-3} - \cdots \\ & - A_1\alpha_{p-1} \wedge \alpha_p + \sum_{m<2p-j-1} b_{j,m}\alpha_j \wedge \alpha_m, \end{aligned}$$

where $\alpha = a_0\alpha_0 + a_1\alpha_1 + \cdots + a_{2p-1}\alpha_{2p-1} + \alpha_{2p}$. Thus

$$(d\alpha)^p = (-1)^p p! A_1 A_2 \cdots A_{p-1}\alpha_0 \wedge \alpha_1 \wedge \alpha_2 \wedge \cdots \wedge \alpha_{2p-1}$$

and $\alpha \wedge (d\alpha)^p \neq 0$. This means that α is a contact form.

Conversely, we suppose now that the Lie algebra $\mathfrak{g}$ admits a contact form α. We put

$$\alpha = b_0\alpha_0 + b_1\alpha_1 + \cdots + b_{2p}\alpha_{2p}.$$

As $Z(\mathfrak{g})$ is not included in $\operatorname{Ker}\alpha$, then $b_{2p} \neq 0$. We have

$$\begin{aligned} d\alpha &= b_0(d\alpha_0) + b_1(d\alpha_1) + \cdots + b_{2p}(d\alpha_{2p}) \\ &= b_2(-\alpha_0 \wedge \alpha_1) + b_3(-\alpha_0 \wedge \alpha_2) + b_4(-\alpha_0 \wedge \alpha_3 - a_{1,4}\alpha_1 \wedge \alpha_2) + \cdots \\ &\quad + b_j\left(\sum_{l+s<j} t_{l,s}\alpha_l \wedge \alpha_s\right) + \cdots \\ &\quad + b_{2p}\biggl(-\alpha_0 \wedge \alpha_{2p-1} - A_{p-1}\alpha_1 \wedge \alpha_{2p-2} - A_{p-2}\alpha_2 \wedge \alpha_{2p-3} - \cdots \\ &\quad - A_1\alpha_{p-1} \wedge \alpha_p + \sum_{m<2p-j-1} b_{j,m}\alpha_j \wedge \alpha_m\biggr). \end{aligned} \tag{$**$}$$

As the basis $(X_0, X_1, \ldots, X_{2p})$ is adapted we have $X_{2p} \in \ker d\alpha_i$, $0 \leqslant i \leqslant 2p$, and thus $X_{2p} \in \ker d\alpha$. But $(d\alpha)^p \neq 0$ and

$$(d\alpha)^p = \lambda\alpha_0 \wedge \alpha_1 \wedge \alpha_2 \wedge \cdots \wedge \alpha_{2p-1}$$

with $\lambda \neq 0$. Let us examine the terms of the expression $(**)$ which appear in the nonzeroproduct $(d\alpha)^p$. In this expression there is only one term containing the form α_{2p-1}; it is the term $-b_{2p}\alpha_0 \wedge \alpha_{2p-1}$. We deduce that

$$(d\alpha)^p = -b_{2p}p\alpha_0 \wedge \alpha_{2p-1} \wedge (d\alpha)^{p-1} = -b_{2p}p\alpha_0 \wedge \alpha_{2p-1} \wedge \theta_1^{p-1}$$

where $\theta_1 \in \wedge^2\mathfrak{g}^*$ is obtained from $d\alpha$ by deleting the terms containing α_0 or α_{2p-1}; thus α_0, α_{2p-1} do not appear in θ_1. Let as examine now α_{2p-2}. In the expression of θ_1 there is only one term containing the form α_{2p-1}; it is the term $-b_{2p}A_{p-1}\alpha_1 \wedge \alpha_{2p-2}$. We deduce that

$$\begin{aligned} (d\alpha)^p &= -b_{2p}p\alpha_0 \wedge \alpha_{2p-1} \wedge \theta_1^{p-1} \\ &= (-b_{2p})^2(p-1)pA_{p-1}(\alpha_0 \wedge \alpha_{2p-1}) \wedge (\alpha_1 \wedge \alpha_{2p-2})\theta_2^{p-2} \end{aligned}$$

where $\theta_2 \in \wedge^2\mathfrak{g}^*$ is obtained from θ_1 by deleting the terms containing α_1 or α_{2p-2}; thus $\alpha_0, \alpha_{2p-1}, \alpha_1, \alpha_{2p-2}$ do not appear in θ_2. By induction we have

$$(d\alpha)^p = (-1)^p p! b_{2p}^p A_1 A_2 \cdots A_{p-1}\alpha_0 \wedge \alpha_1 \wedge \alpha_2 \wedge \cdots \wedge \alpha_{2p-1} \neq 0$$

and $A_1, A_2, \ldots, A_{p-1} \neq 0$. $\square$

2.4. Contact isomorphisms classes of filiform Lie algebras

The following result gives the classification up to contacto-isomorphisms of the contact filiform Lie algebras.

Theorem 4. *Let $\mathfrak{g}$ be a filiform $(2p+1)$-dimensional Lie algebra defined in the basis $(X_0, X_1, \dots, X_{2p})$ by the brackets*

$$[X_0, X_i] = X_{i+1}, \quad i = 1, \dots, 2p-1,$$
$$[X_k, X_{k+1}] = \sum_{r=2k+2}^{2p} a_{k,r} X_r, \quad k = 1, \dots, p-1$$

and let $(\alpha_0, \alpha_1, \dots, \alpha_{2p})$ be the dual basis. If $\alpha = a_0\alpha_0 + a_1\alpha_1 + \dots + a_{2p}\alpha_{2p}$ is a contact form on $\mathfrak{g}$, then the form $\beta = a_{2p}\alpha_{2p}$ is also a contact form on $\mathfrak{g}$ and $(\mathfrak{g}, \alpha)$ is contacto-isomorphic to $(\mathfrak{g}, \beta)$.

Proof. The fact that β is a contact form on $\mathfrak{g}$ is a consequence of the proof of Theorem 3. To prove the theorem it is sufficient to find an automorphism $\varphi \in \operatorname{Aut}\mathfrak{g}$ such that $\varphi^*\alpha = \beta$.

Consider the derivation $d = -a_{2p-1} \operatorname{ad} X_0$. As $\operatorname{ad} X_0(X_i) = X_{i+1}$, $i = 1, \dots, 2p-1$, the automorphism $A = \exp d$ satisfies the following property:

$$A(X_{2p-1}) = X_{2p-1} - a_{2p-1}X_{2p}, \qquad A(X_{2p}) = X_{2p}.$$

Then

$$A^*\alpha_{2p} = \alpha_{2p} - a_{2p-1}\alpha_{2p-1} + \sum_{i<2p-1} c_i\alpha_i$$

and

$$A^*\alpha = a_{2p}\alpha_{2p} + \sum_{j\leqslant 2p-2} q_j\alpha_j.$$

We suppose now that

$$\alpha = a_{2p}\alpha_{2p} + \sum_{j\leqslant 2p-k} b_j\alpha_j, \quad 2 \leqslant k \leqslant 2p-1,$$

and we prove the existence of an automorphism $\varphi \in \operatorname{Aut}\mathfrak{g}$ such that

$$\varphi^*\alpha = a_{2p}\alpha_{2p} + \sum_{j\leqslant 2p-k-1} b'_j\alpha_j.$$

Consider the derivation $d = -\lambda b_{2p-k} \operatorname{ad} X_{k-1}$. By the proof of Theorem 3 we have

$$\operatorname{ad} X_{k-1}(X_{2p-k}) = A_{p-k+1}X_{2p}, \quad 2 \leqslant k \leqslant p,$$

and

$$\operatorname{ad} X_{k-1}(X_{2p-k}) = -A_{k-p}X_{2p}, \quad \text{if } k > p.$$

We have also ad $X_{k-1}(X_i) = 0$ for all indices $i > 2p - k$. Let us put

$$\lambda = \begin{cases} \frac{1}{a_{2p}A_{p-k+1}}, & \text{if } 2 \leqslant k \leqslant p, \\ \frac{-1}{a_{2p}A_{k-p}}, & \text{if } i > 2p-2. \end{cases}$$

Then the automorphism $\varphi = \exp d$ satisfies the required condition. By induction we deduce the theorem. □

Remark 3.

1. The complete list of filiform contact Lie algebras of dimensions $\leqslant 11$ is given in [10].
2. Theorem 4 only concerns the filiform case. But, by direct inspection, we can affirm that this result remains true for every nilpotent Lie algebra of dimension less or equal to 7 [2,24,25].
3. Suppose that α_{2p} is a contact form on the filiform Lie algebra $\mathfrak{g}$. In general the contact Lie algebras $(\mathfrak{g}, a\alpha_{2p})$ and $(\mathfrak{g}, b\alpha_{2p})$ with $a \neq b$ are not contacto-isomorphic. But these contact Lie algebras are always contacto-isomorphic if $a \cdot b > 0$ and $\mathfrak{g}$ admits a semisimple derivation (see [15] and the Section 2.1 for their description).
4. If $K = \mathbb{C}$, then Theorems 2, 3, 4 and the point (2) of this remark are valid. But the point (3) of this remark must be modified (see [10]).

3. Symplectic Lie algebras of dimension $\leqslant 6$

The relation between the classes of symplectic Lie algebras of dimension $2p$ and the contact Lie algebras of dimension $2p + 1$ obtained above and the description of contact structures on filiform Lie algebras and on nilpotent Lie algebras of dimension $\leqslant 7$ permit to obtain some classification results about symplectic Lie algebras. The following theorem gives a complete classification up to symplecto-isomorphism of the symplectic Lie algebras of dimension $\leqslant 6$ (see also [8,11]).

Theorem 5. *Every nilpotent symplectic Lie algebra of dimension* $\leqslant 6$ *is symplecto-isomorphic to one and only one of the following symplectic Lie algebras*:

Dimension 2.

1. $\mathbb{R}^2$,
   $\omega = \alpha_1 \wedge \alpha_2$.

Dimension 4.

1. $[X_1, X_2] = X_3, \quad [X_1, X_3] = X_4$,
   $\omega = \alpha_1 \wedge \alpha_4 + \alpha_2 \wedge \alpha_3$.
2. $[X_1, X_2] = X_3$,
   $\omega = \alpha_1 \wedge \alpha_3 + \alpha_2 \wedge \alpha_4$.
3. $\mathbb{R}^4$,
   $\omega = \alpha_1 \wedge \alpha_4 + \alpha_2 \wedge \alpha_3$.

Dimension 6.

1. $[X_1, X_2] = X_3, \quad [X_1, X_3] = X_4, \quad [X_1, X_4] = X_5,$
   $[X_1, X_5] = X_6, \quad [X_2, X_3] = X_5, \quad [X_2, X_4] = X_6,$
   $\omega = \alpha_1 \wedge \alpha_6 + (1-\lambda)\alpha_2 \wedge \alpha_5 + \lambda\alpha_3 \wedge \alpha_4, \quad \lambda \in \mathbb{R} \setminus \{0, 1\}.$
2. $[X_1, X_2] = X_3, \quad [X_1, X_3] = X_4, \quad [X_1, X_4] = X_5,$
   $[X_1, X_5] = X_6, \quad [X_2, X_3] = X_6,$
   $\omega(\lambda) = \lambda(\alpha_1 \wedge \alpha_6 + \alpha_2 \wedge \alpha_4 + \alpha_3 \wedge \alpha_4 - \alpha_2 \wedge \alpha_5), \quad \lambda \in \mathbb{R} \setminus \{0\}.$
3. $[X_1, X_2] = X_3, \quad [X_1, X_3] = X_4, \quad [X_1, X_4] = X_5, \quad [X_1, X_5] = X_6,$
   $\omega = \alpha_1 \wedge \alpha_6 - \alpha_2 \wedge \alpha_5 + \alpha_3 \wedge \alpha_4.$
4. $[X_1, X_2] = X_3, \quad [X_1, X_3] = X_4, \quad [X_1, X_4] = X_6, \quad [X_2, X_3] = X_5, \quad [X_2, X_5] = X_6,$
   $\omega(\lambda_1, \lambda_2) = \lambda_1\alpha_1 \wedge \alpha_4 + \lambda_2(\alpha_1 \wedge \alpha_5 + \alpha_1 \wedge \alpha_6 + \alpha_2 \wedge \alpha_4 + \alpha_3 \wedge \alpha_5),$
   $\lambda_1 \in \mathbb{R},\ \lambda_2 \in \mathbb{R} \setminus \{0\}.$
5. $[X_1, X_2] = X_3, \quad [X_1, X_3] = X_4, \quad [X_1, X_4] = -X_6,$
   $[X_2, X_3] = X_5, \quad [X_2, X_5] = X_6,$
   $\omega_1(\lambda_1, \lambda_2) = \lambda_1\alpha_1 \wedge \alpha_4 + \lambda_2(\alpha_1 \wedge \alpha_5 + \alpha_1 \wedge \alpha_6 + \alpha_2 \wedge \alpha_4 + \alpha_3 \wedge \alpha_5),$
   $\lambda_1 \in \mathbb{R},\ \lambda_2 \in \mathbb{R} \setminus \{0\},$
   $\omega_2(\lambda) = \lambda(\alpha_1 \wedge \alpha_6 - 2\alpha_1 \wedge \alpha_5 - 2\alpha_2 \wedge \alpha_4 + \alpha_2 \wedge \alpha_6 + \alpha_3 \wedge \alpha_4 + \alpha_3 \wedge \alpha_5), \quad \lambda \in \mathbb{R} \setminus \{0\},$
   $\omega_3(\lambda) = \lambda(\alpha_1 \wedge \alpha_4 - \alpha_1 \wedge \alpha_5 + \alpha_1 \wedge \alpha_6 - \alpha_2 \wedge \alpha_4 + \alpha_2 \wedge \alpha_5$
   $\quad + \alpha_2 \wedge \alpha_6 + \alpha_3 \wedge \alpha_4 + \alpha_3 \wedge \alpha_5), \quad \lambda \in \mathbb{R} \setminus \{0\},$
   $\omega_4(\lambda) = \lambda(2\alpha_1 \wedge \alpha_4 + \alpha_1 \wedge \alpha_6 + 2\alpha_2 \wedge \alpha_5 + \alpha_2 \wedge \alpha_6 + \alpha_3 \wedge \alpha_4 + \alpha_3 \wedge \alpha_5), \quad \lambda \in \mathbb{R} \setminus \{0\}.$
6. $[X_1, X_2] = X_3, \quad [X_1, X_3] = X_4, \quad [X_1, X_4] = X_5, \quad [X_2, X_3] = X_6,$
   $\omega_1 = \alpha_1 \wedge \alpha_6 + \alpha_2 \wedge \alpha_4 + \alpha_2 \wedge \alpha_5 - \alpha_3 \wedge \alpha_4,$
   $\omega_2 = -\alpha_1 \wedge \alpha_6 - \alpha_2 \wedge \alpha_4 - \alpha_2 \wedge \alpha_5 + \alpha_3 \wedge \alpha_4.$
7. $[X_1, X_2] = X_4, \quad [X_1, X_4] = X_5, \quad [X_1, X_5] = X_6,$
   $[X_2, X_3] = X_6, \quad [X_2, X_4] = X_6,$
   $\omega_1(\lambda) = \lambda(\alpha_1 \wedge \alpha_3 + \alpha_2 \wedge \alpha_6 - \alpha_4 \wedge \alpha_5), \quad \lambda \in \mathbb{R} \setminus \{0\},$
   $\omega_2(\lambda) = \alpha_1 \wedge \alpha_6 + \alpha_2 \wedge \alpha_5 - \alpha_3 \wedge \alpha_4, \quad \lambda \in \mathbb{R} \setminus \{0\},$
   $\omega_3(\lambda) = -\alpha_1 \wedge \alpha_6 - \alpha_2 \wedge \alpha_5 + \alpha_3 \wedge \alpha_4, \quad \lambda \in \mathbb{R} \setminus \{0\}.$
8. $[X_1, X_3] = X_4, \quad [X_1, X_4] = X_5, \quad [X_1, X_5] = X_6,$
   $[X_2, X_3] = X_5, \quad [X_2, X_4] = X_6,$
   $\omega = \alpha_1 \wedge \alpha_6 + \alpha_2 \wedge \alpha_5 - \alpha_3 \wedge \alpha_4.$
9. $[X_1, X_2] = X_4, \quad [X_1, X_4] = X_5, \quad [X_1, X_5] = X_6, \quad [X_2, X_3] = X_6,$
   $\omega(\lambda) = \lambda(\alpha_1 \wedge \alpha_3 + \alpha_2 \wedge \alpha_6 - \alpha_4 \wedge \alpha_5), \quad \lambda \in \mathbb{R}\setminus\{0\}.$
10. $[X_1, X_2] = X_4, \quad [X_1, X_4] = X_5, \quad [X_1, X_3] = X_6, \quad [X_2, X_4] = X_6,$
    $\omega_1 = \alpha_1 \wedge \alpha_6 + \alpha_2 \wedge \alpha_5 - \alpha_2 \wedge \alpha_6 - \alpha_3 \wedge \alpha_4,$
    $\omega_2 = -\alpha_1 \wedge \alpha_6 - \alpha_2 \wedge \alpha_5 + \alpha_2 \wedge \alpha_6 + \alpha_3 \wedge \alpha_4.$
11. $[X_1, X_2] = X_4, \quad [X_1, X_4] = X_5,$
    $[X_2, X_3] = X_6, \quad [X_2, X_4] = X_6,$
    $\omega_1(\lambda) = \alpha_1 \wedge \alpha_6 + \alpha_2 \wedge \alpha_5 + \lambda\alpha_2 \wedge \alpha_6 - \alpha_3 \wedge \alpha_4, \quad \lambda \in \mathbb{R},$
    $\omega_2(\lambda) = -\alpha_1 \wedge \alpha_6 - \alpha_2 \wedge \alpha_5 + \lambda\alpha_2 \wedge \alpha_6 + \alpha_3 \wedge \alpha_4, \quad \lambda \in \mathbb{R}.$

12. $[X_1,X_2]=X_4,\quad [X_1,X_4]=X_5,\quad [X_1,X_3]=X_6,$
$[X_2,X_3]=-X_5,\quad [X_2,X_4]=X_6,$
$\omega(\lambda)=\lambda\alpha_1\wedge\alpha_5+\alpha_2\wedge\alpha_6+(\lambda+1)\alpha_3\wedge\alpha_4,\quad \lambda\in\mathbb{R}\setminus\{0,-1\}.$

13. $[X_1,X_2]=X_4,\quad [X_1,X_3]=X_5,\quad [X_1,X_4]=X_6,$
$[X_2,X_3]=X_6,$
$\omega_1(\lambda)=\alpha_1\wedge\alpha_6+\lambda\alpha_2\wedge\alpha_5+(\lambda-1)\alpha_3\wedge\alpha_4,\quad \lambda\in\mathbb{R}\setminus\{0,1\},$
$\omega_2(\lambda)=\alpha_1\wedge\alpha_6+\lambda\alpha_2\wedge\alpha_4+\alpha_2\wedge\alpha_5+\alpha_3\wedge\alpha_5,\quad \lambda\in\mathbb{R}\setminus\{0\},$
$\omega_3=\alpha_1\wedge\alpha_6+\alpha_2\wedge\alpha_4+\frac{1}{2}\alpha_2\wedge\alpha_5-\frac{1}{2}\alpha_3\wedge\alpha_4.$

14. $[X_1,X_2]=X_4,\quad [X_1,X_4]=X_6,\quad [X_1,X_3]=X_5,$
$\omega_1=\alpha_1\wedge\alpha_6+\alpha_2\wedge\alpha_4+\alpha_3\wedge\alpha_5,$
$\omega_2=\alpha_1\wedge\alpha_6-\alpha_2\wedge\alpha_4+\alpha_3\wedge\alpha_5,$
$\omega_3=\alpha_1\wedge\alpha_6+\alpha_2\wedge\alpha_5+\alpha_3\wedge\alpha_4.$

15. $[X_1,X_2]=X_4,\quad [X_1,X_4]=X_6,\quad [X_2,X_3]=X_5,$
$\omega_1=-\alpha_1\wedge\alpha_5+\alpha_1\wedge\alpha_6+\alpha_2\wedge\alpha_5+\alpha_3\wedge\alpha_4,$
$\omega_2=\alpha_1\wedge\alpha_5-\alpha_1\wedge\alpha_6-\alpha_2\wedge\alpha_5-\alpha_3\wedge\alpha_4,$
$\omega_3=\alpha_1\wedge\alpha_6+\alpha_2\wedge\alpha_4+\alpha_3\wedge\alpha_5.$

16. $[X_1,X_2]=X_5,\quad [X_1,X_3]=X_6,$
$[X_2,X_4]=X_6,\quad [X_3,X_4]=-X_5,$
$\omega_1=\alpha_1\wedge\alpha_6+\alpha_2\wedge\alpha_3-\alpha_4\wedge\alpha_5,$
$\omega_2=\alpha_1\wedge\alpha_6-\alpha_2\wedge\alpha_3+\alpha_4\wedge\alpha_5.$

17. $[X_1,X_3]=X_5,\quad [X_1,X_4]=X_6,\quad [X_2,X_3]=X_6,$
$\omega=\alpha_1\wedge\alpha_6+\alpha_2\wedge\alpha_5+\alpha_3\wedge\alpha_4.$

18. $[X_1,X_2]=X_4,\quad [X_1,X_3]=X_5,\quad [X_2,X_3]=X_6,$
$\omega_1(\lambda)=\alpha_1\wedge\alpha_6+\lambda\alpha_2\wedge\alpha_5+(\lambda-1)\alpha_3\wedge\alpha_4,\quad \lambda\in\mathbb{R}\setminus\{0,1\},$
$\omega_2(\lambda)=\alpha_1\wedge\alpha_5+\lambda\alpha_1\wedge\alpha_6-\lambda\alpha_2\wedge\alpha_5+\alpha_2\wedge\alpha_6-2\lambda\alpha_3\wedge\alpha_4,\quad \lambda\in\mathbb{R}\setminus\{0\},$
$\omega_3=\alpha_3\wedge\alpha_5-\alpha_1\wedge\alpha_6+\alpha_2\wedge\alpha_5+2\alpha_3\wedge\alpha_4.$

19. $[X_1,X_2]=X_4,\quad [X_1,X_4]=X_5,\quad [X_1,X_5]=X_6,$
$\omega=\alpha_1\wedge\alpha_3+\alpha_2\wedge\alpha_6-\alpha_4\wedge\alpha_5.$

20. $[X_1,X_2]=X_3,\quad [X_1,X_3]=X_4,$
$[X_1,X_4]=X_5,\quad [X_2,X_3]=X_5,$
$\omega=\alpha_1\wedge\alpha_6+\alpha_2\wedge\alpha_5-\alpha_3\wedge\alpha_4.$

21. $[X_1,X_2]=X_4,\quad [X_1,X_4]=X_6,\quad [X_2,X_3]=X_6,$
$\omega_1=\alpha_1\wedge\alpha_6+\alpha_2\wedge\alpha_4-\alpha_3\wedge\alpha_4-\alpha_3\wedge\alpha_5,$
$\omega_2=\alpha_1\wedge\alpha_6+\alpha_2\wedge\alpha_5-\alpha_3\wedge\alpha_4,$
$\omega_3=-\alpha_1\wedge\alpha_6-\alpha_2\wedge\alpha_5+\alpha_3\wedge\alpha_4.$

22. $[X_1,X_2]=X_5,\quad [X_1,X_5]=X_6,$
$\omega=\alpha_1\wedge\alpha_6+\alpha_2\wedge\alpha_5+\alpha_3\wedge\alpha_4.$

23. $[X_1,X_2]=X_5,\quad [X_1,X_3]=X_6,$
$\omega_1=\alpha_1\wedge\alpha_6+\alpha_2\wedge\alpha_5+\alpha_3\wedge\alpha_4,$
$\omega_2=\alpha_1\wedge\alpha_4+\alpha_2\wedge\alpha_6+\alpha_3\wedge\alpha_5,$
$\omega_3=\alpha_1\wedge\alpha_4+\alpha_2\wedge\alpha_6-\alpha_3\wedge\alpha_5.$

24. $[X_1, X_4] = X_6, \quad [X_2, X_3] = X_5,$
$\omega_1 = \alpha_1 \wedge \alpha_6 + \alpha_2 \wedge \alpha_5 + \alpha_3 \wedge \alpha_4,$
$\omega_1 = -\alpha_1 \wedge \alpha_6 - \alpha_2 \wedge \alpha_5 - \alpha_3 \wedge \alpha_4.$
25. $[X_1, X_2] = X_6, \quad \omega = \alpha_1 \wedge \alpha_6 + \alpha_2 \wedge \alpha_5 + \alpha_3 \wedge \alpha_4.$
26. $\mathbb{R}^6, \quad \omega = \alpha_1 \wedge \alpha_6 + \alpha_2 \wedge \alpha_5 + \alpha_3 \wedge \alpha_4.$

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